

EARLY-WARNING NETWORK FOR LLM DRIFT

SEISMOGRAPH

A federated, privacy-preserving early-warning network for silent behavioural drift in third-party LLM & agent APIs.

“Is it me, my prompt, or did the model silently change underneath me?”

Open Source · Apache-2.0

107 tests · CI-gated

Live public dashboard

`pip install seismograph-probe`

The model changes underneath you

Providers silently re-route, re-quantise, or re-tune deployed endpoints. The API stays stable, the response still returns 200 OK — but the behaviour moves, and that drift flows straight into every product built on top.

Conventional observability cannot see it. It watches infrastructure, not meaning:

Latency

✓ normal

Error rate

✓ normal

Uptime

✓ normal

Model behaviour

✗ drifted

Three green dashboards, one silent regression.

For six weeks, no one could tell

A documented case: the Anthropic Claude 3.5 Sonnet degradation, Aug–Sep 2025. A load-balancer misconfiguration misrouted traffic to an incompatible model. The diagnostic gap between onset and disclosure:



The question SEISMOGRAPH answers: could an instrument have caught this on day one?

The solution: federated canary probes

1

Probe

A fixed, content-addressed canary suite (≤ 200 prompts at temperature 0) is replayed against the target model on a schedule. Same stimulus, compared to itself over time.

2

Privatise

Raw prompts and outputs are destroyed at the probe boundary. Only SHA-256 hashes and differentially-private distributional features ever leave the perimeter.

3

Correlate

Signals from many independent observers are scored together. A drift alert fires only when ≥ 2 distinct observers agree — never on one org's noise.

Correlation-first by design: a single observer can never raise a public alarm.

Proof: detected 38 days early

38

days before the
postmortem

19

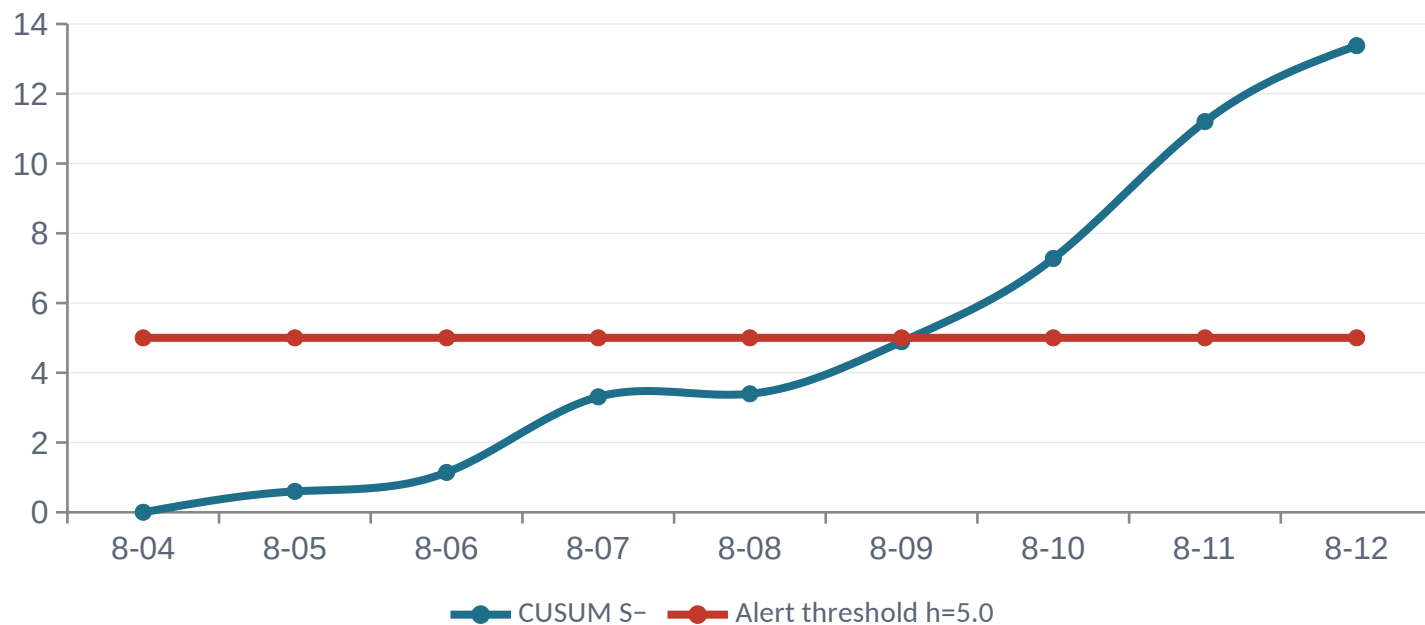
days before user-
visible escalation

0.8%

traffic affected
at detection

1

probe — quorum
scales the signal



Reading the signal

The CUSUM detector crossed its alert threshold on 10 Aug 2025 — inside the subtle 0.8% window, weeks before the fault was visible to anyone.

$S^- = 7.28$ vs $h = 5.0 \cdot \mu_0 = 0.990, \sigma_0 = 0.0044$

Reproducible from a fixed seed.

Privacy is the moat

Cross-organisation correlation only works if no one has to expose their prompts. SEISMOGRAPH makes that structural, not a policy promise.

Hashing

Canary identifiers cross the wire only as non-reversible SHA-256 digests.
Raw text never leaves the probe.

Differential privacy

Calibrated Laplace noise at $\epsilon = 2.0$ per flush. A budget accountant enforces a rolling 24-hour limit and sleeps the probe before overspend.

Pseudonymous federation

Each probe holds an Ed25519 key. The engine knows public keys and reputation — never org identity. Fabricated signals start at zero trust.

Raw prompts and outputs are destroyed at the aggregation boundary — by construction, verified on every change.

Why now

1

Every team is exposed

Production systems increasingly depend on third-party LLM/agent endpoints they neither host nor control.

2

No incumbent

Observability vendors watch latency, tokens and uptime. None watch semantic behaviour across providers.

3

The standard exists

OpenTelemetry GenAI conventions (gen_ai.*, mcp.*) make vendor-neutral instrumentation possible for the first time.

The bet

Silent drift in shared model infrastructure is a systemic risk. The defence is a shared, privacy-preserving early-warning network — a public good first, an enterprise product second.

“Model weather” for everyone who ships on top of an LLM.

Built and working — not a deck-only idea

107

tests passing,
CI-gated

Apache-2.0

open source
licence

Live

public dashboard
on Render

PyPI

pip install
seismograph-probe

Component	Status
CUSUM change-point detector	live
Privacy / DP aggregation layer	live
Ingestion gateway + signed schema	live
Cross-observer agreement scorer	live
Webhooks & alerting	live
Ed25519 verification · OTel emission	hardening

We state maturity honestly: live components are verified; hardening items are in progress for Phase 2.

Roadmap

0	Validation	Reproducible backtest proving detection lead-time vs a documented incident.	COMPLETE
1	Public good	OSS probe + single-node ingestion + free public “model weather” dashboard.	CORE COMPLETE
2	Network growth	DP hardening, OTel/MCP adapters, full Sybil resistance, ClickHouse, methodology paper.	IN PROGRESS
3	Enterprise	Multi-tenant, SSO/RBAC, SOC 2, in-VPC probe, SLAs, canary-gated rollback.	PLANNED

From public good to product

Public good

Free

Public “model weather” dashboard for the top model tuples. Drives adoption and the observer network.

Team

Private fleet

Your own probes, private drift history, webhooks & alerting across the models you depend on.

Enterprise

In-VPC + SLA

In-VPC probe, SSO/RBAC, SOC 2, canary-gated rollback, support SLAs for regulated teams.

Adoption compounds the moat: every new observer makes every drift alert more trustworthy.

The ask

Phase 0 is validated and the public good is live. We are raising support to fund Phase 2 — the network that turns a working probe into a shared early-warning system.

- **Grant funding** to resource DP hardening, OTel/MCP adapters and the methodology paper (Phase 2).
- **Design partners** teams running on third-party LLMs who will host a probe and corroborate live drift.
- **Researchers & contributors** on change-point detection, differential privacy and federated trust.